Tidy data, joining and project organization

Data Science Flipped Classroom - Module 7

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Tidy data

The same data, many shapes

The same information can be stored in many different table layouts.

Most of them make analysis harder than it needs to be.

Tidy data is one consistent layout that fits the tidyverse, so `dplyr`, `ggplot2`, and the rest all work without reshaping first.

Data semantics

Before we can say what makes a dataset tidy, we need a shared vocabulary for what is in it. A dataset is a collection of values, each one a number or a string.

Every value belongs to both a variable and an observation:

- A variable holds all the values that measure the same attribute (height, temperature, duration) across units
- An observation holds all the values measured on the same unit (a person, a day, a country-year) across attributes

The three rules

- Each variable forms a column
- Each observation forms a row
- Each cell is a single measurement

> If your table follows these three rules, it is tidy. Almost everything else this module does is about getting there.

Tidy in practice

`tidyr` ships with the same dataset in several shapes. This one is tidy:

```r
table1
```

```
# A tibble: 6 × 4
  country      year  cases population
  <chr>       <dbl>  <dbl>      <dbl>
1 Afghanistan  1999    745   19987071
2 Afghanistan  2000   2666   20595360
3 Brazil       1999  37737  172006362
4 Brazil       2000  80488  174504898
5 China        1999 212258 1272915272
6 China        2000 213766 1280428583
```

One row per country-year, one column per variable: `cases` and `population`.